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Department of Computer Engineering (S.E.)

Project Based Learning Report

Project Name : Sentiment Analysis Using BERT

PBL Batch : C-6

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Project Guide :

Abstract

This project focuses on the implementation of a sentiment analysis system using the BERT model, a state-of-the-art transformer-based architecture. The aim is to classify text inputs based on the emotional tone conveyed—such as positive, negative, or neutral sentiment. Although this project uses the IMDB movie reviews dataset as a case study, the methodology is designed to be generic and applicable across domains such as product reviews, social media posts, customer feedback, and news articles.

Introduction

Sentiment analysis, also known as opinion mining, is a natural language processing (NLP) technique used to determine the emotional tone behind a body of text. It is widely used in business, marketing, and social media monitoring to analyze customer feedback, product reviews, and public sentiment.

The traditional approaches to sentiment analysis include rule-based systems and machine learning models like Naive Bayes, SVMs, and logistic regression. However, these methods often struggle with understanding context and sarcasm. With the advent of transformer-based models like BERT, it is now possible to capture deep contextual meaning, significantly improving performance on NLP tasks.

Problem Statement

The goal of this project is to build a sentiment analysis model using the BERT architecture to classify user-generated textual content—such as reviews, comments, or social media posts—into sentiment categories like positive, negative, or neutral. The model is designed to be adaptable to various domains, improving real-time sentiment detection and supporting decision-making systems in diverse applications.

Dataset

- Name: IMDB Movie Review Dataset
- Description: Contains 50,000 movie reviews with labels indicating positive or negative sentiment.
- Split:
 - 60% for training
 - 30% for testing
 - 10% for validation

Preprocessing Steps:

- Removal of special characters and unnecessary punctuation
- Tokenization using BertTokenizer
- Padding and truncation to fit fixed input size of BERT

Model and Architecture

- Model Used: BERT (from HuggingFace Transformers library)
- Tokenizer: BertTokenizer - Converts text into tokens suitable for the BERT model
- Architecture:
 - Tokenization of input text
 - BERT base model generates embeddings
 - Dropout layer to prevent overfitting
 - Linear layer for classification
 - Softmax activation for output probabilities

Special Tokens:

- [CLS] - Used to represent the entire sentence
- [SEP] - Separates segments in case of pair inputs

Why BERT?

- Pre-trained on massive corpora
- Bidirectional context understanding
- Fine-tuning improves performance on specific tasks

Training Details

- Framework: PyTorch
- Programming Language: Python
- Libraries/Resources Used:
 - Transformers (HuggingFace)
 - PyTorch
 - scikit-learn
 - numpy
 - pandas
 - tqdm
 - requests
 - zipfile
- Optimizer: AdamW (Adam optimizer with weight decay)
- Learning Rate Scheduler: Linear schedule with warmup using `get_linear_schedule_with_warmup`
- Loss Function: CrossEntropyLoss
- Epochs: 4
- Batch Size: 32
- Learning Rate: $2e-5$

Data Handling:

- `train_test_split` from scikit-learn used to create training and validation sets
- `DataLoader` used to batch and shuffle data during training

Evaluation

- Metrics:
 - Accuracy
 - Precision
 - Recall
 - F1 Score

Model Evaluation Approach:

- Evaluate predictions against test set
- Generate classification report
- Use confusion matrix for error analysis

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Results

- Accuracy: ~92% on test data
- Observations:
 - BERT significantly outperforms traditional models
 - Handles contextual nuances better
 - Performs well on short and long reviews

Saving and Deployment

- Model saved using `torch.save()` as .pt file
- Can be loaded later for inference
- Deployment options include REST APIs using Flask or FastAPI

Challenges Faced

- Long training time on CPU
- Managing input sequence length for BERT
- Overfitting with small datasets

Solutions:

- Used GPU for faster training
- Applied regularization and dropout
- Fine-tuned learning rate and batch size

Future Scope

- Extend to multi-class sentiment (e.g., neutral, mixed)
- Use multilingual BERT for non-English datasets
- Integrate into chatbots or customer feedback systems
- Real-time sentiment prediction for social media streams
- Apply to political opinion analysis, brand monitoring, healthcare feedback

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Conclusion

The project successfully demonstrates the power of transformer-based models like BERT for sentiment analysis. Through fine-tuning and careful data preparation, the model achieves high accuracy and generalizes well to unseen data. With growing text data online, such models have significant practical value in opinion mining and decision-making systems.

References

- [Hugging Face Transformers Documentation](#)
- [PyTorch Documentation](#)
- [IMDB Dataset on Kaggle](#)